

AMBER: Banked Embedding Compute as a Verifiable Offline Knowledge Artifact

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Amber by Teresa

Reference implementation: github.com/offchainthoughts/amber

Abstract

We present AMBER, a format and protocol that turn the one-time cost of embedding a corpus into a single portable artifact that is a cryptographic commitment to the exact corpus and embedder that produced it. The artifact is the image of a document set under a pinned, deterministic embedding map, and the committed root is a public, regenerable witness to that map. Trust in the banked compute is gated by two independently verifiable predicates. An integrity check recomputes the commitment in $O(n)$ hashing and needs no model. A probabilistic *audit* re-embeds a random sample of $k \ll n$ chunks and shows that the stored vectors are the faithful image of the source under the pinned map. We formalize the embedder as a deterministic map quantized to a fixed-point lattice, give the commitment with explicit domain separation, and prove a sequence of propositions that bound an adversary’s advantage by the sample size and the tamper fraction. We separate what the artifact binds, namely corpus identity and embedding fidelity, from what it does not. It carries no claim about any downstream generator, and we prove the intuitive reading that “the artifact *is* the answer” void. We state the guarantee against an explicit hierarchy of adversary classes and identify the exact anchor at which corpus authenticity, as opposed to embedding authenticity, becomes external.

1 Introduction

Embedding a corpus is a bank-once, use-many computation. The heavy pass over documents runs once, and the resulting vectors are reused for offline retrieval. The banked result is portable and works without a network; only a light runtime remains at task time. A portable artifact that one did not produce oneself raises a question the retrieval literature rarely asks. Were the banked vectors honestly computed? A consumer holding a stored index has no model loaded and cannot, in general, tell faithful embeddings from fabricated ones.

AMBER composes the two so that the banked artifact is a structural commitment to the computation that produced it, not just a container of vectors. The design objective is a single invariant.

(I) Trust in a banked artifact requires the conjunction of an integrity witness over its committed bytes and a faithful-embedding witness sampled from the source. A self-consistent artifact alone is insufficient, and either witness alone is insufficient for the adversary classes of §7.

Realizing (I) needs care, because the intuitive reading, that the artifact *is* the knowledge, is provably void (Proposition 3). Our contribution is a construction that delivers verifiable, reproducible, offline-portable banked compute, reduces its guarantee to standard hash assumptions, and gives a precise account of the boundary of that guarantee.

1.1 Notation and primitives

We write $\parallel$ for byte concatenation, H for SHA-256, and $|x|$ for byte length. A corpus $\mathcal{D}$ is an ordered set of sources. An embedder $E : \Sigma^* \rightarrow \mathbb{R}^d$ maps text to a unit vector; Q denotes fixed-point quantization to $\mathbb{Z}_{[-127,127]}^d$ and Q^{-1} its dequantization. A chunker C deterministically segments $\mathcal{D}$ into an ordered list of chunks. We write $\text{negl}(\lambda)$ for a negligible function of the security parameter λ , here the hash output length.

2 System overview

AMBER is stratified into three layers with minimal coupling, summarized in Table 1. The corpus is public and regenerable; the bank is the frozen embedding compute; the commitment is a succinct public root that binds the two. The decomposition is the source of the security argument: authenticity is confined to two scalars, the commitment root r and an external corpus anchor $r_{\mathcal{D}}$.

Layer	Object	Cost to verify
Corpus	chunks $c_i = C(\mathcal{D})_i$	external anchor
Bank	vectors $q_i = Q(E(c_i))$	$O(k)$ re-embed
Commitment	root r over (c_i, q_i)	$O(n)$ or $O(\log n)$

Table 1: Layer decomposition. No layer substitutes for another. Integrity binds Bank to Commitment; audit binds Bank to Corpus.

3 The chunker and embedder

Construction 1 (Deterministic chunker). *Normalize each source by NFC and whitespace collapse, then emit fixed word-count windows of size w with overlap $o < w$, in source order. The parameters (w, o, norm) are serialized canonically and hashed; the digest enters the manifest, so two builders with the same corpus and parameters emit byte-identical chunks.*

Construction 2 (Pinned embedder). *The embedder’s identity and configuration are serialized to a canonical descriptor whose digest is committed. The reference default is a signed feature-hashing map: tokens are hashed (keyed BLAKE2B) to a bucket and sign, term frequencies accumulate, and the vector is ℓ_2 -normalized. It is dependency-free and bitwise reproducible. A semantic transformer is a drop-in under Construction 3.*

4 The commitment

Definition 1 (Leaf). *For chunk c_i and quantized vector q_i in a sequence of length n , the leaf payload is $p_i = \text{canon}(c_i) \parallel 0x1f \parallel q_i$, and the leaf hash binds position and length,*

$$\ell_i = H(0x00 \parallel i \parallel n \parallel |p_i| \parallel p_i).$$

Internal nodes use a distinct tag $0x01$, so a leaf can never be reinterpreted as a node. The root $r = \text{MR}(\ell_0, \dots, \ell_{n-1})$ is the commitment. A separate root $r_{\mathcal{D}}$ over $\text{canon}(c_i)$ alone fixes corpus identity independently of the embedder.

Definition 2 (Artifact). *An artifact is the tuple $A = (\text{manifest}, (c_i), (q_i))$ serialized as a single file, where the manifest carries the embedder and chunker digests, d , n , r , and $r_{\mathcal{D}}$.*

5 Reproducibility under quantization

Floating-point embedding is not guaranteed bitwise-stable across hardware, so a naive commitment to raw floats would be unverifiable in practice. We commit to quantized vectors and verify within a tolerance.

Construction 3 (τ -reproducibility). *E is τ -reproducible if for the pinned configuration and any chunk c , two evaluations q, q' satisfy $\|q - q'\|_{\infty} \leq \tau$. The deterministic default achieves $\tau = 0$; a pinned transformer revision achieves a small τ absorbed by Q .*

6 Verification and audit

Proposition 1 (Integrity). *If the stored (c_i, q_i) differ in any byte from those committed by r , then recomputing the root yields $r' \neq r$ except with probability $\text{negl}(\lambda)$ under collision resistance of H .*

Proof sketch. A differing artifact that recomputes to the same root yields, by the standard Merkle extractor walking from root to the changed leaf, a collision of H on tagged, length-bound inputs. Domain separation prevents leaf and node confusion. $\square$

Integrity is cheap and model-free, but it does not establish that the q_i were honestly produced by E . A forger may commit self-consistent fabricated vectors. The audit closes this gap.

Protocol 1 (Audit). *Sample k leaf indices uniformly without replacement using OS entropy. For each sampled i , recompute $\hat{q}_i = Q(E(c_i))$ and declare leaf i inconsistent if $\|\hat{q}_i - q_i\|_{\infty} > \tau$. Reject if any sampled leaf is inconsistent.*

Proposition 2 (Audit soundness). *Let m of the n leaves be inconsistent with the pinned E at tolerance τ , and let $\rho = m/n$. The audit rejects with probability*

$$1 - \frac{\binom{n-m}{k}}{\binom{n}{k}} \geq 1 - (1 - \rho)^k,$$

independent of which leaves are corrupt. To bound a missed tamper fraction ρ at confidence $1 - \delta$ it is enough to take $k \geq \lceil \ln(1/\delta) / \rho \rceil$ approximately.

Proof sketch. The probability that a without-replacement sample of size k avoids all m inconsistent leaves is $\binom{n-m}{k} / \binom{n}{k}$. Sampling without replacement only increases detection relative to the independent bound $(1 - \rho)^k$. $\square$

Remark 1. Audit cost is $O(k)$ embeddings with $k \ll n$, giving trust in arbitrarily large banked compute at sublinear marginal cost. The bound weakens gracefully as ρ shrinks and is vacuous for a single-leaf forgery, which is the price of sampling.

7 What the artifact does and does not bind

Proposition 3 (Scope; the “artifact is the answer” is void). *The commitment binds (a) corpus identity via $r_{\mathcal{D}}$ and (b) em-*

bedding fidelity under E via r and the audit. It binds no property of any downstream generator. For any two generators $G \neq G'$ consuming identical retrieved context, G and G' may emit different outputs, so no function of r fixes a task output. Banked retrieval is therefore not banked reasoning.

We state the guarantee against an explicit hierarchy.

$\mathcal{A}_1$ **transport tamperer** flips bytes of a stored artifact. Defeated by Proposition 1.

$\mathcal{A}_2$ **lazy forger** commits self-consistent vectors without running E . Defeated by Proposition 2, not by integrity.

$\mathcal{A}_3$ **corpus substituter** swaps source text but re-embeds honestly. Detected by comparing $r_{\mathcal{D}}$ to an independent anchor.

Out of scope an adversary who honestly runs E on a corpus of their choosing. AMBER authenticates the map, not the truthfulness of the inputs. This is the exact boundary at which an external $r_{\mathcal{D}}$ anchor, such as a signed release or a transparency log, becomes necessary.

8 Related work

Retrieval-augmented generation with offline vector stores is well established and is not claimed here as novel. AMBER adds a verifiable, reproducible, portable commitment layer over it. Merkle trees give the integrity primitive; vector commitments and authenticated data structures give the position binding. The audit is a sampling argument in the spirit of proofs of retrievability and provable data possession, specialized to the predicate that the stored vector is the pinned map’s image of the stored chunk. Content-addressed datasets and data-version tools hash artifacts but do not audit the computation relating two committed layers, which is the gap Proposition 2 fills.

9 Limitations

The default embedder is lexical, not semantic; retrieval quality needs a pinned transformer, which trades $\tau = 0$ for a small tolerance and a revision pin. The audit assumes the auditor can run E . It certifies fidelity to a map, not the quality or truthfulness of the corpus. Corpus authenticity needs an external anchor. Quantization is lossy, and inputs near cell boundaries can inflate τ . None of these affect Propositions 1, 2, and 3, which concern binding rather than retrieval quality.

References

- [1] R. C. Merkle. A digital signature based on a conventional encryption function. *CRYPTO*, 1987.
- [2] P. Lewis et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*, 2020.
- [3] A. Juels and B. Kaliski. PORs: proofs of retrievability for large files. *ACM CCS*, 2007.
- [4] G. Ateniese et al. Provable data possession at untrusted stores. *ACM CCS*, 2007.
- [5] D. Catalano and D. Fiore. Vector commitments and their applications. *PKC*, 2013.
- [6] K. Weinberger et al. Feature hashing for large scale multi-task learning. *ICML*, 2009.